

Project Proposal: Milestone 4

Proposal for Analyzing Life Expectancy Factors: A Regression Approach

Introduction:

Life expectancy is a crucial indicator of a population's overall health and well-being, reflecting the effectiveness of healthcare systems, socio-economic conditions, and lifestyle factors. This proposal aims to analyze factors influencing life expectancy using data sourced from the World Health Organization (WHO) and supplemented by the United Nations and other reputable sources. By employing regression modeling techniques, including Artificial Neural Networks (ANN) and Gradient Booster Regression, this study seeks to uncover the relationships between various predictors and life expectancy, providing insights that can lead to great implications for policymakers and healthcare practitioners among other authorities.

Problem Statement

The objective of this project is to identify and quantify the factors that significantly influence life expectancy across different countries. Understanding these factors can help policymakers develop targeted interventions to improve population health outcomes.

Importance of the Problem

Life expectancy is not only a measure of a country's health status but also a reflection of its socio-economic development. Identifying the key determinants of life expectancy can aid in formulating strategies to address health disparities, allocate resources efficiently, and improve overall quality of life.

Target Audience

This project is aimed at government officials, policymakers, and international organizations such as the WHO and the United Nations. These stakeholders can leverage the insights from this study to design and implement effective health policies and interventions.

Data Source

The data was originally collected from WHO and United Nations websites, with contributions from Deeksha Russell and Duan Wang. The dataset was compiled and updated by Lasha Gochiashvili and made available on Kaggle. It comprises comprehensive health-related factors for 193 countries, collected over the period from 2000 to 2015.

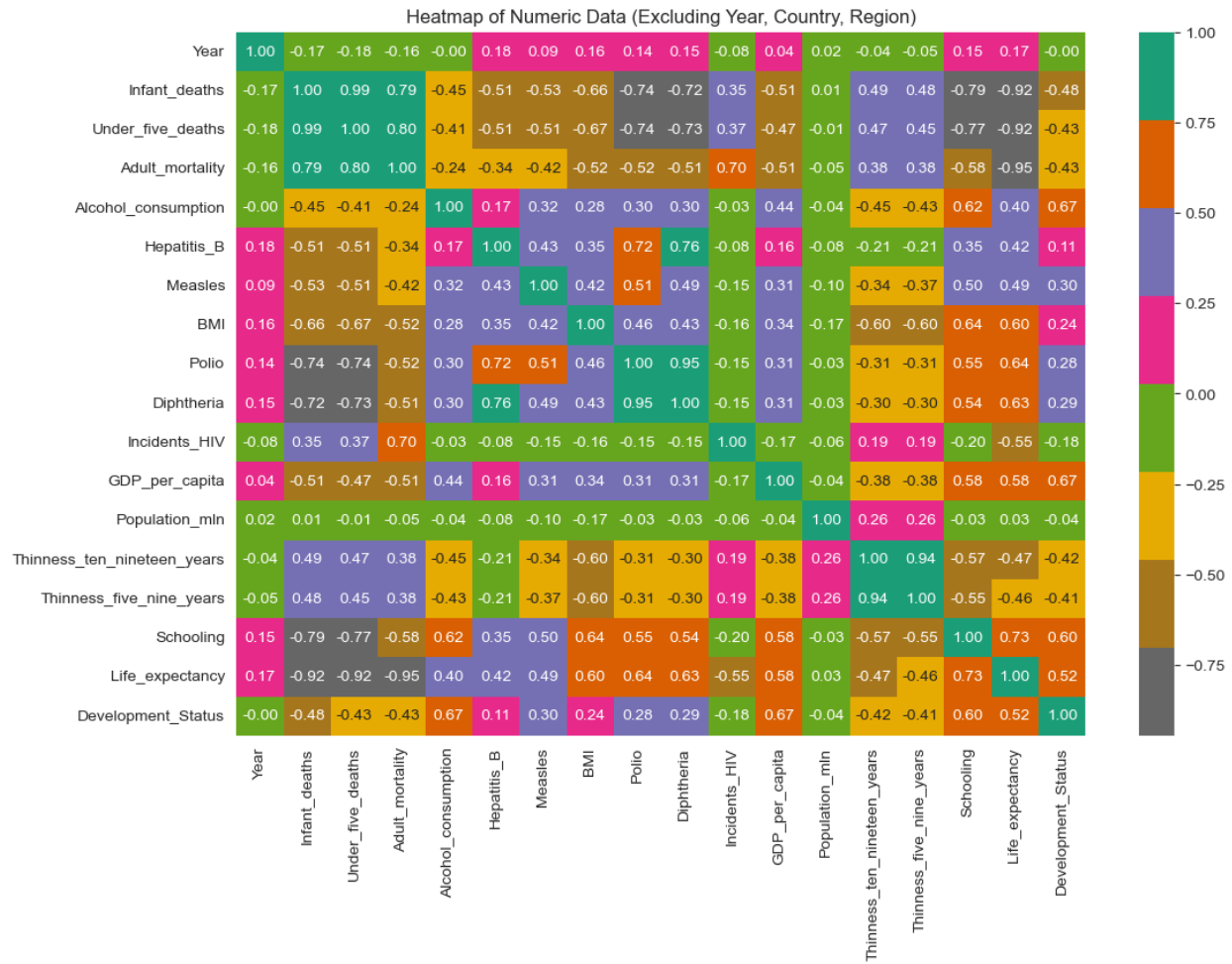
Relevance of the Data

The dataset includes a wide range of variables related to health, socio-economic status, and environmental factors, making it suitable for a detailed analysis of life expectancy determinants. The richness and diversity of the data allow for a robust examination of the interplay between different factors affecting life expectancy.

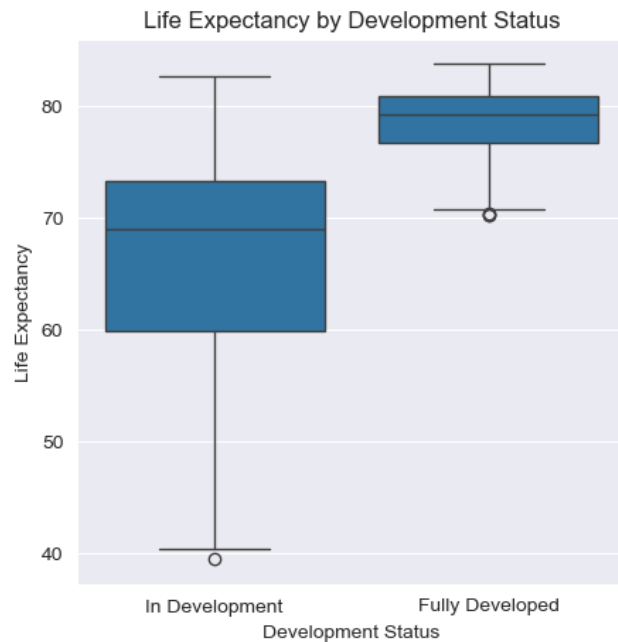
Limited Snapshot of the Dataset

	Country	Region	Year	Infant_deaths	Under_five_deaths	Adult_mortality	Alcohol_consumption	Hepatitis_B	Measles	BMI
0	Turkiye	Middle East	2015	11.1	13.0	105.8240	1.32	97	65	27.8
1	Spain	European Union	2015	2.7	3.3	57.9025	10.35	97	94	26.0
2	India	Asia	2007	51.5	67.9	201.0765	1.57	60	35	21.2
3	Guyana	South America	2006	32.8	40.5	222.1965	5.68	93	74	25.3
4	Israel	Middle East	2012	3.4	4.3	57.9510	2.89	97	89	27.0

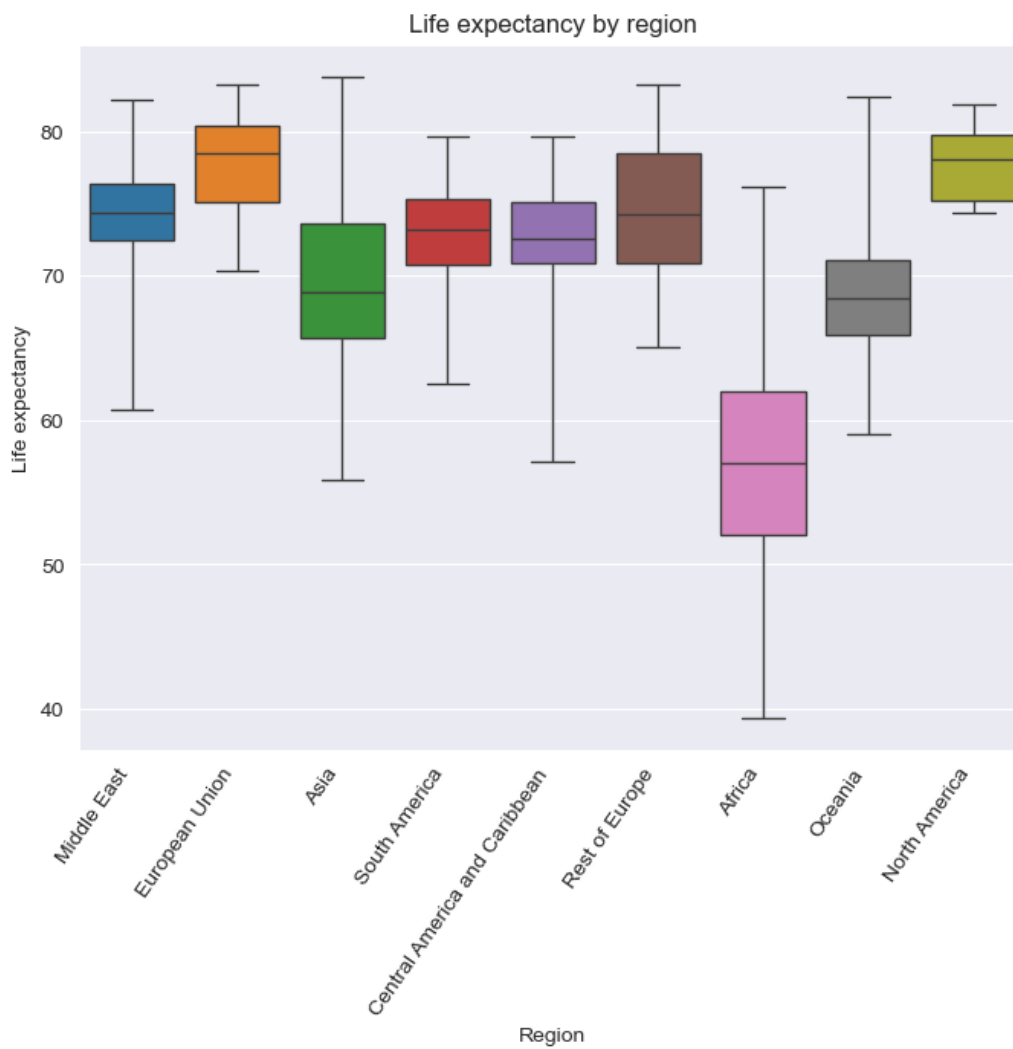
Heatmap showing how that variables correlate to each other and to life_expectancy



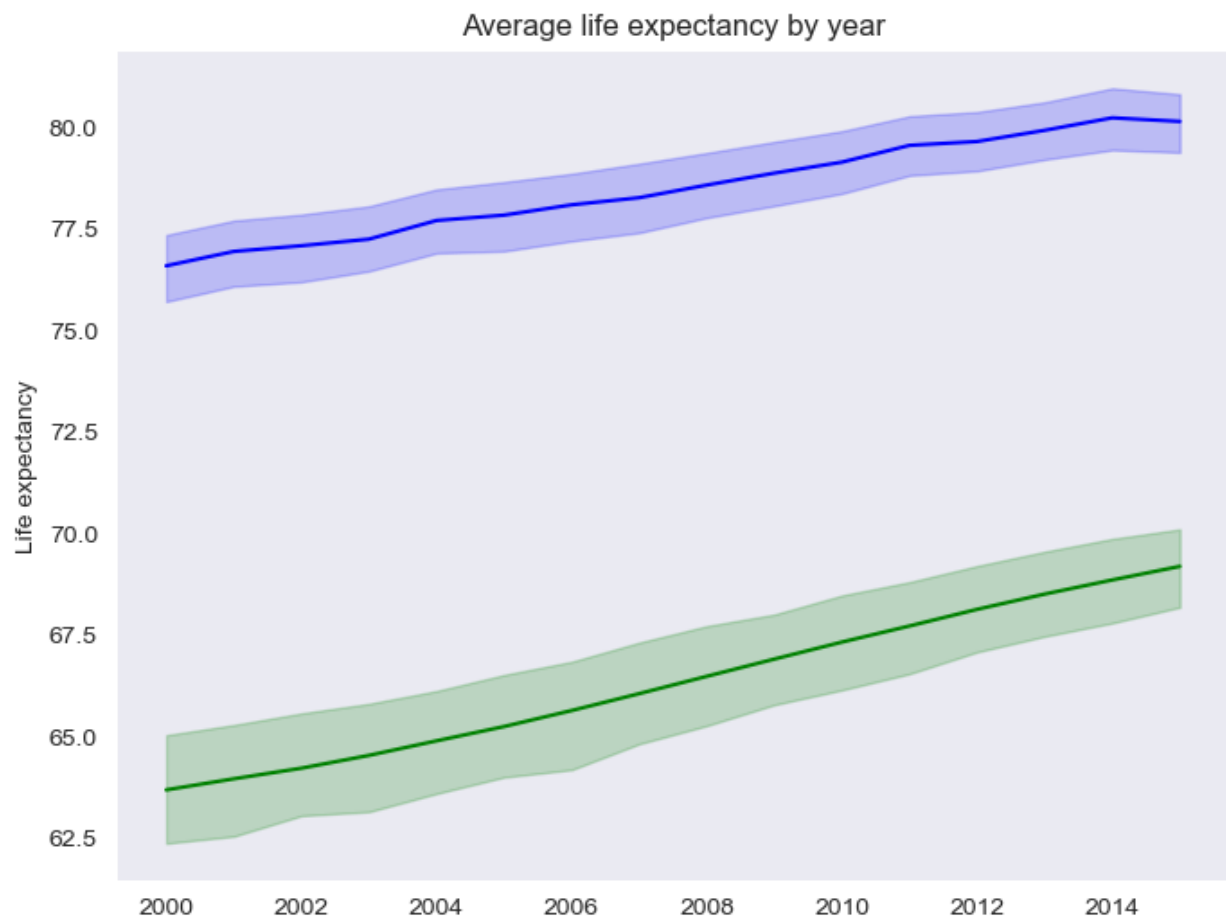
Boxplot highlighting the variability in Life expectancy based on the development status of the countries.



Boxplot that showcases the life expectancy by Region.



Historical line plot where we can see a positive trend of life expectancy.



Methods

Data Exploration and Preparation

Initial exploration of the data revealed significant correlations between life expectancy and variables such as GDP per capita, schooling, and vaccination coverage for diseases like Polio, Diphtheria, and Hepatitis B. Scatterplots and heatmaps were instrumental in uncovering these relationships.

Data preparation steps included:

- Handling missing values by imputing with mean values.
- Encoding categorical variables: The categorical variable 'Development_Status' was encoded using label encoding to convert it into numerical format.
- Standardizing features using standard scaling to ensure uniformity.

Modeling Approaches

Two regression models were employed:

1. Gradient Boosting Regression: Known for its robustness and ability to handle heterogeneous data, this model combines multiple weak learners to improve prediction accuracy.
2. Artificial Neural Networks (ANN): Chosen for their ability to capture complex nonlinear relationships among predictors and life expectancy.

Evaluation Metrics

The models were evaluated using the following metrics:

- Mean Absolute Error (MAE): Measures the average magnitude of errors in predictions.
- Root Mean Square Error (RMSE): Provides a measure of the differences between predicted and actual values.
- R-squared: Indicates the proportion of variance in the dependent variable explained by the model.

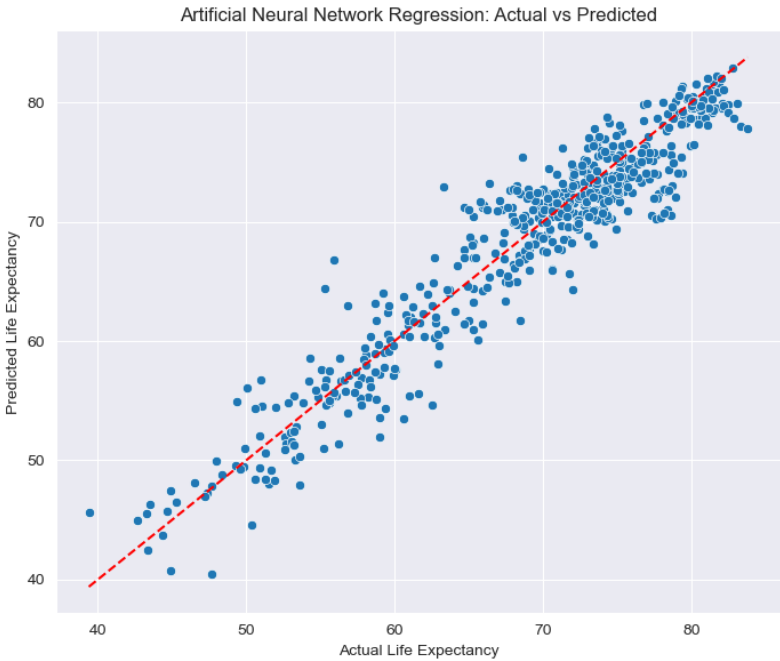
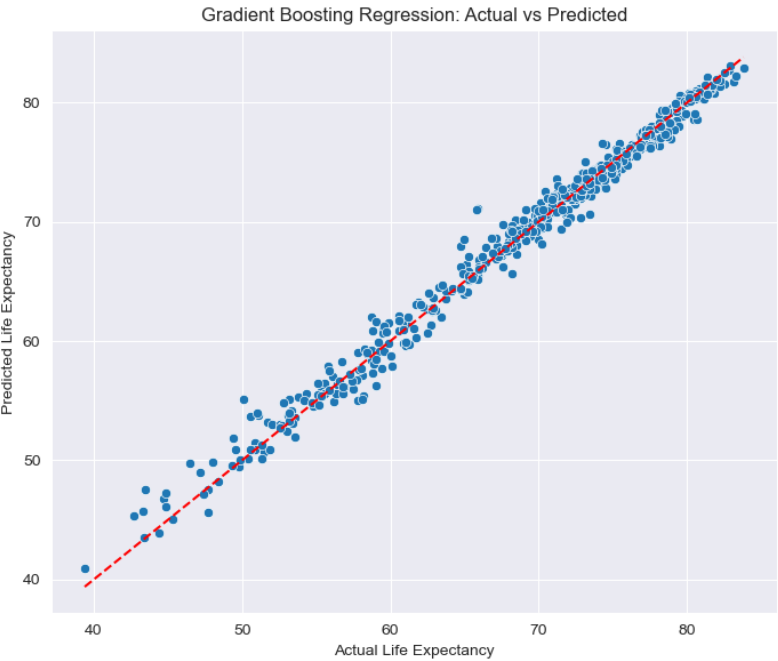
Cross-validation techniques were used to ensure the generalizability of the models and mitigate overfitting risks. The cross-validated MAE was 1.56, indicating the model's performance on unseen data.

Results

Gradient Boosting Regression(optimized):	Artificial Neural Network:
Training MAE: 0.34	Training MAE: 1.919
Testing MAE: 0.74	Testing MAE: 2.042
Training RMSE: 0.46	Training RMSE: 2.495
Testing RMSE: 1.05	Testing RMSE: 2.658
Training R-squared: 0.998	Training R-squared: 0.931
Testing R-squared: 0.987	Testing R-squared: 0.915

Visualizations

Scatterplots comparing actual vs predicted values for both models highlighted the accuracy of predictions. These visualizations confirmed the models' ability to capture the relationships between the predictors and life expectancy.



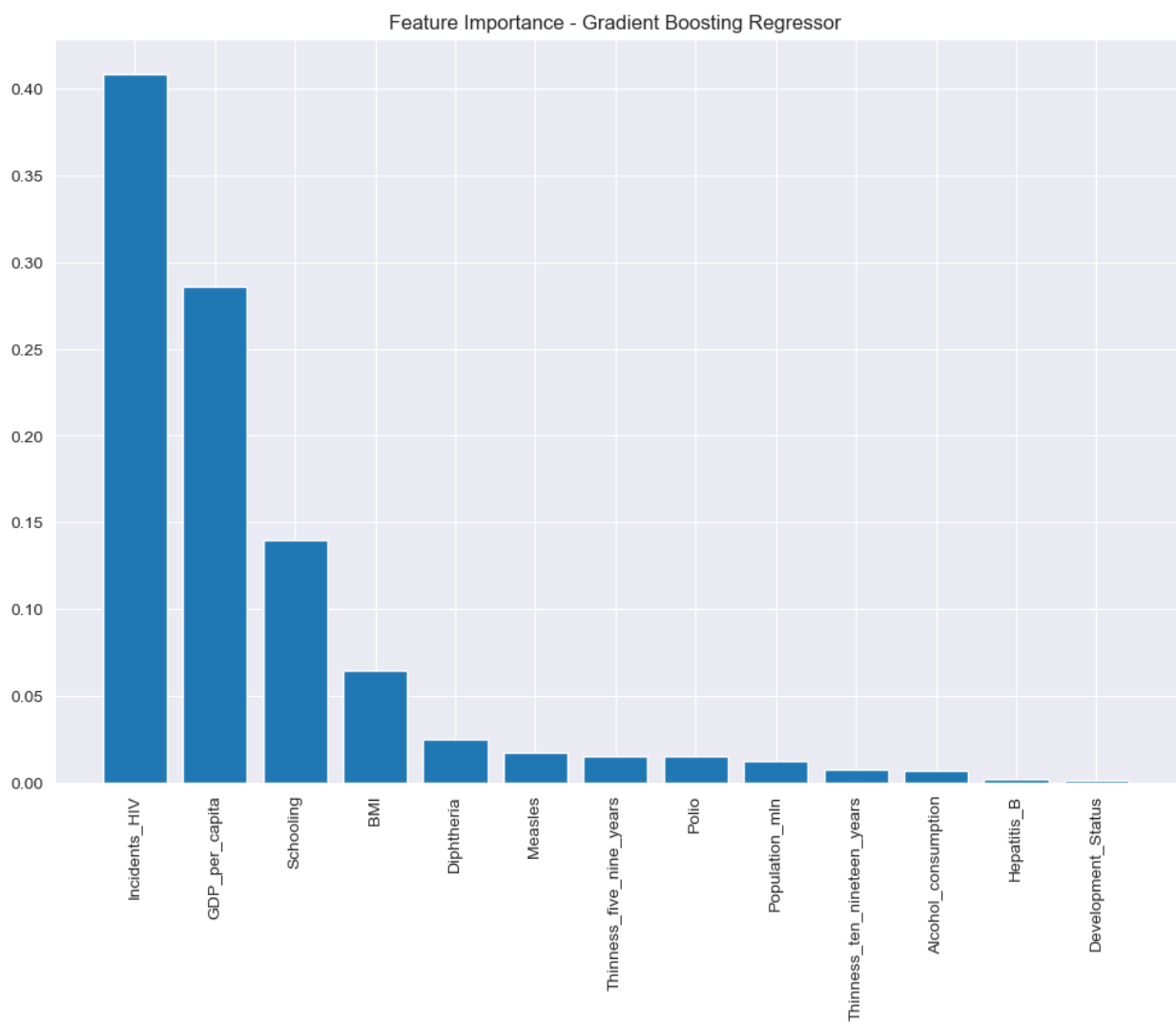
Interpretation Comparison between models:

- MAE and RMSE: The Gradient Boosting Regression model outperforms the Artificial Neural Network Regression model in terms of both MAE and RMSE on both the training and testing datasets. The Gradient Boosting model has significantly lower errors, indicating better accuracy in predicting life expectancy.
- R-squared: Similarly, the Gradient Boosting Regression model achieves higher R-squared values, indicating better fit to the data compared to the Artificial Neural Network Regression model.
- Overall, the results suggest that the Gradient Boosting Regression model is superior to the Artificial Neural Network Regression model for this prediction task. It achieves higher accuracy and better fits the data, making it a more suitable choice for predicting life expectancy.

Conclusion

Learnings and Recommendations

The analysis revealed that socio-economic factors such as GDP per capita, schooling, and vaccination coverage are strong predictors of life expectancy. These findings underscore the importance of economic development and robust healthcare infrastructure in improving population health.



Recommendations:

- **Socio-Economic Development:** Policies aimed at improving education and economic conditions can have a significant impact on life expectancy.
- **Healthcare Infrastructure:** Investments in healthcare systems and adequate healthcare infrastructure, including vaccination programs and access to healthcare services, is essential for improving life expectancy, particularly in countries that are still under development.
- **Targeted Interventions:** Tailored interventions addressing specific health challenges, such as vaccination campaigns for preventable diseases and initiatives to address malnutrition, can contribute to improvements in life expectancy. Using these models as guide to focus the analysis by specific countries to further analyze what features are more relevant to each country.

Model Readiness and Future Work

While the models demonstrate high accuracy, further validation with additional data is recommended. Future work should explore region-specific models to account for local variations in health determinants.

Ethical Considerations:

Ethical considerations include ensuring data privacy, addressing potential biases in the data, and interpreting results responsibly. Transparency in methodology and acknowledging limitations are crucial to maintain the integrity of the analysis.

Continued Monitoring and Evaluation:

Regular monitoring and evaluation of health outcomes, supported by predictive modeling techniques like Gradient Boosting Regression, are essential for identifying emerging health trends and assessing the effectiveness of interventions. Policymakers and healthcare practitioners should prioritize data-driven decision-making and evidence-based interventions to improve population health outcomes.

In conclusion, the analysis highlights the importance of addressing socio-economic disparities, strengthening healthcare systems, and implementing targeted interventions to improve population health and increase life expectancy. By focusing on these areas, policymakers, healthcare departments and other government authorities can work towards achieving better health outcomes and enhancing overall well-being for populations worldwide by focusing the resources and efforts towards efficient and strongly influential features.

References:

Gochiashvili, L. (2023). Life Expectancy (WHO) Fixed [Data set]. Kaggle.

<https://doi.org/10.34740/KAGGLE/DS/3065197>

World Health Organization. (n.d.). Global Health Observatory (GHO) Data Repository.

Retrieved from [https://www.who.int/data/gho/data/indicators/indicator-details/GHO/life-expectancy-at-birth-\(years\)](https://www.who.int/data/gho/data/indicators/indicator-details/GHO/life-expectancy-at-birth-(years))

United Nations. (n.d.). World Bank Data. Retrieved from <https://data.worldbank.org/>

Our World in Data. (n.d.). Mean Years of Schooling. Retrieved from

<https://ourworldindata.org/grapher/mean-years-of-schooling-long-run>